React – Fundamentals

Session – 7

1. Create a React component called TrendingSongs that uses the useEffect hook to log 'Component mounted' in the console when the component first renders.

Ans.

TrendingSongs:

```
import { useEffect } from "react";

function TrendingSongs() {
  useEffect(() => {
    console.log("Component mounted");
  }, []);

  return (
    <div>
      <h2>Trending Songs</h2>
    </div>
  );
}

export default TrendingSongs;
```

2. Build a React component called IPLScoreFetcher that uses useEffect to fetch live match data from https://jsonplaceholder.typicode.com/posts when the component mounts, and display the title of the first post as the current 'match headline'.\

Ans.

```
import { useEffect, useState } from "react";

function IPLScoreFetcher() {
  const [headline, setHeadline] = useState("");

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/posts")
      .then((response) => response.json())
      .then((data) => {
        setHeadline(data[0].title);
      })
      .catch((error) => console.error(error));
  }, []);

  return (
```

```
    <div>
      <h2>Current Match Headline</h2>
      <p>{headline}</p>
    </div>
  );
}

export default IPLScoreFetcher;
```

3. Create a MovieSuggestions component that fetches a list of movies from https://jsonplaceholder.typicode.com/users inside useEffect on mount, and displays the names in a list. Show a loading message while the data is being fetched.

Hint: Use a loading state variable and update it once data is received.

Ans.

```
import { useEffect, useState } from "react";

function MovieSuggestions() {
  const [movies, setMovies] = useState([]);
  const [loading, setLoading] = useState(true);

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/users")
      .then((response) => response.json())
      .then((data) => {
        setMovies(data);
        setLoading(false);
      })
      .catch((error) => {
        console.error(error);
        setLoading(false);
      });
  }, []);

  if (loading) {
    return <p>Loading...</p>;
  }

  return (
    <div>
      <h2>Movie Suggestions</h2>
      <ul>
        {movies.map((movie) => (
          <li key={movie.id}>{movie.name}</li>
        ))}
      </ul>
```

```
    </div>
  );
}

export default MovieSuggestions;
```

4. Refactor an existing React component that fetches data in a button click handler to instead use useEffect so the data is fetched automatically when the component mounts, not on button click.

Constraint: Only move the fetch logic; do not change the UI or styling.

Ans.

```
import { useState, useEffect } from "react";

function DataComponent() {
  const [data, setData] = useState([]);

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/posts")
      .then((response) => response.json())
      .then((data) => setData(data))
      .catch((error) => console.error(error));
  }, []);

  return (
    <div>
      <button>Load Data</button>

      <ul>
        {data.map((item) => (
          <li key={item.id}>{item.title}</li>
        ))}
      </ul>
    </div>
  );
}

export default DataComponent;
```